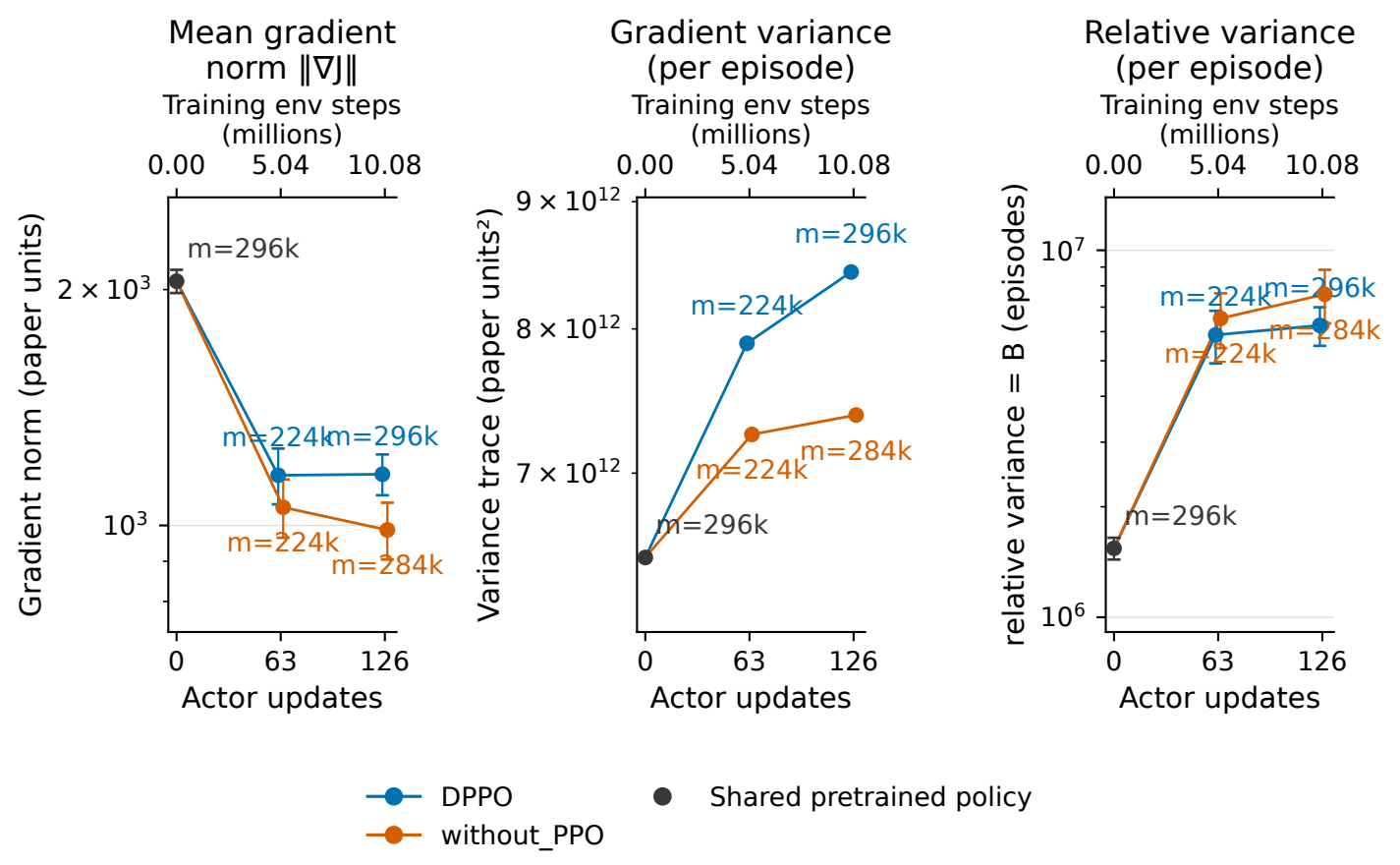
Mean gradient norm ‖∇J‖
Training env steps (millions)
0.00
5.04
10.08
Gradient norm (paper units)
2 × 10^3
10^3
m=296k
m=224k
m=296k
m=224k
m=284k
0
63
126
Actor updates
Gradient variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
Variance trace (paper units²)
9 × 10^12
8 × 10^12
7 × 10^12
m=296k
m=224k
m=284k
m=224k
m=296k
0
63
126
Actor updates
Relative variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
relative variance = B (episodes)
10^7
10^6
m=224k
m=296k
m=284k
m=224k
m=296k
0
63
126
Actor updates
DPPO
without_PPO
Shared pretrained policy